

NAOMAOHU, China

WMO: 521120

Lat: 43.7537N

Lon: 94.9748E

Elev: 480

StdP: 95.69

Time Zone: 8.00 (E08)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.5	(c) -19.0	(d) -29.4	(e) 0.3	(f) -14.3	(g) -27.2	(h) 0.3	(i) -12.9	(j) 9.3	(k) -6.9	(l) 8.0	(m) -7.9	(n) 1.6	(o) 250	(p) 0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.3	(c) 39.5	(d) 18.6	(e) 38.1	(f) 18.1	(g) 36.7	(h) 17.6	(i) 19.8	(j) 34.8	(k) 19.1	(l) 33.9	(m) 18.6	(n) 33.3	(o) 3.4	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	11.8	23.9	13.9	10.5	25.5	12.2	9.4	26.9	58.9	34.9	56.6	33.9	54.6	33.1	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	11.1	9.4	-22.6	42.6	3.3	1.1	-25.0	43.4	-26.9	44.1	-28.8	44.7	-31.1	45.5
			-23.1	20.7	3.1	0.9	-25.3	21.4	-27.1	21.9	-28.8	22.4	-31.1	23.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	-10.6	-4.1	6.6	16.2	21.9	28.3	30.5	28.3	21.7	12.1	1.4	-8.9
	DBStd	14.88	4.37	4.77	5.14	5.07	3.83	3.28	2.80	2.97	3.92	4.35	5.04	4.77
	HDD10.0	2052	639	396	132	12	0	0	0	0	0	28	259	586
	HDD18.3	3564	898	629	365	96	13	1	0	0	14	197	507	844
	CDD10.0	2791	0	0	26	198	369	548	634	569	352	93	2	0
	CDD18.3	1261	0	0	0	32	124	298	376	311	116	3	0	0
	CDH23.3	16592	0	0	1	364	1429	4050	5495	4058	1158	37	0	0
CDH26.7	9092	0	0	0	98	608	2306	3356	2267	453	4	0	0	
Wind	WSAvg	3.7	2.3	3.0	3.9	5.0	5.0	4.6	4.2	3.9	3.7	3.2	2.8	2.3
Precipitation	PrecAvg	83	1	1	5	4	7	16	21	12	7	4	2	1
	PrecMax	135	3	4	37	11	48	39	50	38	17	16	7	8
	PrecMin	57	0	0	0	1	1	2	5	1	1	0	0	0
	PrecStd	22	1	1	9	3	10	9	11	7	4	4	2	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	10.3	22.6	32.2	36.8	40.6	42.1	40.4	35.2	26.6	16.3	4.1
		MCWB	-1.7	2.2	9.1	14.1	15.9	18.6	19.9	18.9	16.0	12.3	7.4	-0.9
	2%	DB	-0.2	7.3	20.3	29.3	34.1	38.6	40.2	38.2	33.0	24.1	13.4	1.4
		MCWB	-4.0	0.7	8.0	12.6	15.1	17.8	19.2	18.1	15.7	11.1	5.7	-2.5
	5%	DB	-2.3	5.3	18.1	27.3	32.3	37.1	38.7	36.7	31.2	22.0	10.8	-0.4
		MCWB	-5.5	-0.6	7.1	12.0	14.5	17.3	18.6	17.7	14.9	10.2	4.2	-3.8
	10%	DB	-4.4	3.4	15.8	25.3	30.3	35.7	37.1	35.2	29.4	19.7	8.6	-2.4
		MCWB	-7.1	-1.8	5.8	10.9	13.6	16.8	18.1	17.3	14.1	9.1	2.7	-5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.6	2.6	9.7	14.6	16.6	19.9	21.0	20.4	17.3	13.1	8.0	-0.4
		MCDB	2.2	9.8	21.3	30.2	33.1	33.9	36.9	34.7	31.9	24.5	14.8	3.5
	2%	WB	-3.7	0.8	8.4	13.3	15.7	19.0	20.1	19.4	16.3	11.8	6.0	-2.3
		MCDB	-0.4	7.1	19.3	27.5	32.5	33.8	35.6	34.0	30.3	22.5	13.1	1.3
	5%	WB	-5.4	-0.5	7.3	12.3	14.9	18.3	19.5	18.6	15.5	10.6	4.4	-3.7
		MCDB	-2.4	5.0	17.4	26.0	30.9	33.4	34.6	33.0	29.3	20.9	10.6	-0.5
	10%	WB	-7.1	-1.7	6.1	11.3	14.1	17.7	18.9	18.0	14.6	9.4	2.9	-5.2
		MCDB	-4.5	3.1	15.4	24.5	28.9	32.9	33.7	32.3	28.2	19.3	8.3	-2.7
Mean Daily Temperature Range	5% DB	MDBR	9.9	11.6	13.5	12.8	12.3	11.5	11.3	11.8	12.7	12.5	10.0	8.8
		MCDBR	10.6	13.1	15.2	15.1	15.2	13.9	13.2	13.7	14.5	14.6	12.0	9.7
		MCWBR	7.8	8.4	8.0	7.0	6.8	5.3	4.9	5.3	6.0	7.2	7.0	6.9
	5% WB	MCDBR	10.5	12.8	14.4	13.4	13.3	11.8	11.5	11.5	13.2	13.2	11.7	9.5
		MCWBR	7.8	8.2	7.6	6.5	6.1	4.4	4.0	4.5	5.5	6.7	6.9	6.9
Clear-Sky Solar Irradiance	taub	0.316	0.353	0.393	0.437	0.442	0.461	0.456	0.426	0.392	0.366	0.337	0.315	
	taud	2.221	2.159	2.065	2.009	2.059	2.095	2.151	2.217	2.244	2.231	2.241	2.223	
	Ebn at Noon	788	818	831	825	833	817	816	828	827	795	755	746	
	Edh at Noon	90	115	144	166	162	157	147	133	120	106	88	82	
All-Sky Solar Radiation	RadAvg	2.21	3.28	4.80	6.07	7.08	7.03	6.86	6.28	5.32	3.87	2.50	1.88	
	RadStd	0.12	0.16	0.21	0.31	0.33	0.42	0.28	0.26	0.19	0.11	0.15	0.15	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (3 neighbors)		+0.46	N/A	N/A	+0.65	-1.02	-1.46	N/A	N/A	+146	+101	

Nomenclature: See separate page